Development of a RASSOR Rover Remote Operation System with AI-Driven Digital Twin for Latency Compensation

AS-IS Report

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Executive Summary

Remote robotic exploration plays a critical role in advancing scientific discovery and supporting future space missions. However, planetary rover operations are significantly constrained by communication latency between Earth and extraterrestrial environments. In the case of lunar exploration, the approximate three-second one-way communication delay creates a six-second round-trip delay between operator commands and rover feedback. This delay disrupts the human-rover control loop, reducing operator situational awareness and increasing the risk of navigation errors, obstacle collisions, and inefficient mission performance.

This project, conducted in collaboration with the Florida Space Institute (FSI), focuses on developing and evaluating a remotely operated RASSOR rover system integrated with a predictive digital twin framework. The purpose of the project is to investigate how predictive modeling and real-time environmental sensing can mitigate the operational challenges caused by communication latency during remote rover teleoperation. The system integrates RGB-D sensing through an OAK-D Lite camera with a digital twin developed in a simulation environment, enabling the system to estimate the rover's current state and predict short-term motion during communication delays. By providing operators with predictive visual feedback, the system aims to improve navigation accuracy, reduce collision frequency, and enhance overall mission performance.

The project follows the Lean Six Sigma DMADV (Define, Measure, Analyze, Design, Verify) methodology to guide system development and evaluation. During the Define phase, the team established the problem context, project objectives, and operational constraints associated with latency-affected rover control. The Measure phase focuses on collecting baseline performance data using controlled testing and simulation environments to evaluate key performance metrics, including response latency, path deviation, obstacle collisions, and task completion time. The Analyze phase examines the relationship between system inputs, such as communication delay, sensor accuracy, and rover speed-and performance outcomes. These analyses inform the Design phase, where multiple system alternatives were evaluated, including enhanced manual teleoperation, digital twin-assisted control, and AI-enhanced navigation. Based on feasibility and effectiveness, a predictive digital twin-assisted system was selected as the preferred design solution.

The project is currently approximately halfway complete. Early milestones, including project planning, proposal development, budgeting, and initial presentations, have been successfully completed. The team is presently focused on finalizing the physical rover platform and integrating electronic components, including onboard processing hardware and sensor systems. Minor delays occurred during hardware assembly due to extended 3D printing times and temporary material shortages; however, these issues were mitigated through schedule adjustments and parallel progress in documentation and system planning. Upcoming work includes testing multiple wheel configurations, collecting experimental data under simulated communication delays, and developing the digital twin environment for predictive modeling.

The expected outcome of this project is a functional rover prototype capable of remote teleoperation under simulated lunar latency conditions, supported by a synchronized digital twin framework. The system will demonstrate how predictive modeling can improve operator awareness and reduce the negative impacts of delayed communication. Results from the project will provide the Florida Space Institute with a prototype architecture, performance data, and documentation that can support future research into latency-resilient robotic systems for space exploration.

Beyond its immediate application to lunar rover operations, the project contributes to broader advancements in digital twin technology, robotics, and cyber-physical systems. These technologies have potential applications not only in space exploration but also in industries such as manufacturing, disaster response, and autonomous systems. By improving the reliability and efficiency of remote robotic operation, this research supports the development of more resilient and sustainable exploration systems for future missions beyond Earth.

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1. Introduction

Space exploration continues to expand scientific knowledge and technological capabilities while introducing complex operational challenges that limit the effectiveness of current systems. One of the most significant challenges is communication latency in remote robotic operations. Due to the distance between Earth and the Moon, delays in transmitting commands and receiving feedback disrupt control and reduce operator situational awareness. Even short delays can negatively impact navigation accuracy, system safety, and overall mission performance, particularly in dynamic and uncertain environments.

To address this challenge, this project focuses on the development and evaluation of a remotely operated RASSOR rover system integrated with a predictive digital twin. The system operates under simulated conditions with communication delay, replicating constraints experienced in lunar environments. By combining RGB-D sensing for environmental data acquisition with predictive modeling through a digital twin, the system is designed to enhance operator decision-making and mitigate the negative effects of delayed feedback. This approach supports the Florida Space Institute (FSI) in advancing rover capabilities and exploring more reliable methods for remote robotic operation.

The purpose of this report is to document and evaluate the current state of the rover system and its associated processes. This includes identifying existing performance limitations, understanding system constraints, and establishing measurable baselines for key performance metrics such as navigation accuracy, task completion time, and collision frequency. The results of this baseline analysis provide a foundation for identifying performance gaps and guiding future system improvements.

This project follows the Lean Six Sigma DMADV (Define, Measure, Analyze, Design, Verify) methodology, which is appropriate for developing and refining a new system rather than improving an existing one. The Define and Measure phases establish the problem context and characterize current system performance, while the Analyze phase identifies key drivers of inefficiency. The Design and Verify phases will focus on developing and validating improved system solutions, particularly through the integration of the digital twin. This structured methodology ensures that engineering design decisions are data-driven, systematically evaluated, and aligned with project objectives.

The process improvement team consists of multidisciplinary members from Industrial Engineering and Data Science, including:

- Jennifer Cifuentes (Team Lead)
- Noah Choi
- Sean Crites
- Matteo D’Agostino
- Mattias Escudero
- Jonathan Gyure
- Abdiel Joseph

The project is supported by Lean Six Sigma mentors, including Mr. Ahmed Mohammed, Dr. Richard Biehl, and Dr. Ahmad Elshennawy, who provides guidance on methodology, analysis, and project development.

From an engineering design perspective, the project operates under several constraints that influence system development and implementation. These include technical constraints such as communication latency and hardware limitations, operational constraints related to system integration and real-time performance, and resource constraints including time, budget, and available equipment. Additionally, safety, reliability, and maintainability considerations are incorporated to ensure the system can operate effectively in simulated mission conditions and support future scalability.

The remainder of this document is organized as follows. Section 2 presents the Define phase, including the problem statement, project objectives, and scope of the system under study. Section 3 describes the Measure phase, where current system performance is quantified using collected data and appropriate analysis tools. Section 4 details the Analyze phase, which examines the relationships between key variables and identifies the primary causes of performance limitations. Section 5 introduces preliminary design alternatives developed based on the analysis results. Section 6 provides a risk analysis of the proposed approaches. Section 7 summarizes the findings and key insights from the project thus far. Section 8 discusses broader impacts related to sustainability, safety, and societal considerations. Finally, Section 9 presents the current status of the project, including the project schedule and progress to date.

2. DMADV - Define:

The process under examination is remote teleoperation of planetary rovers under communication latency conditions, specifically within Earth-to-Moon operations. Traditional teleoperation systems are designed for near real-time feedback, but lunar missions introduce an inherent three second 1 way delay, summing to a 6 second two way delay in total system feedback as shown below. This delay leads to:

- Reduced situational awareness
- Delayed reaction to hazards
- Increased navigational errors
- Elevated risk of hardware damage and mission inefficiency

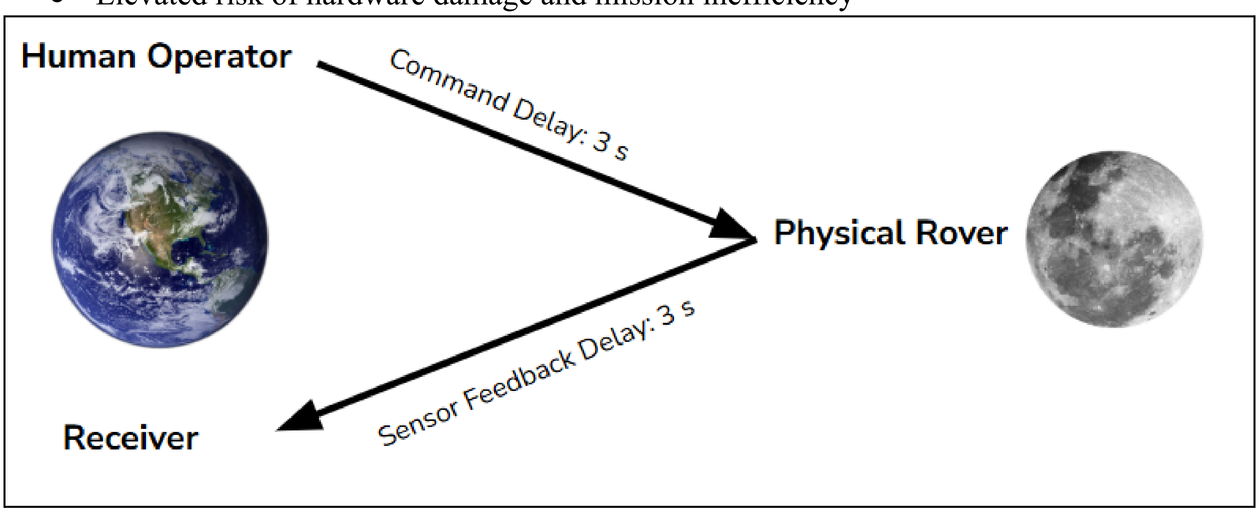


Figure 1: Earth to Moon delay visualization

This project introduces a predictive digital twin framework integrated with RGB-D sensing into a RASSOR rover system, shifting the process from reactive teleoperation to predictive-assisted teleoperation.

2.1 Problem Statement

Remote operation of lunar and planetary rovers is fundamentally constrained by communication latency rather than limitations in rover mobility. The approximate three-second one-way delay between Earth and the Moon results in a six-second round-trip communication lag, causing operators to rely on outdated sensor and visual data when making control decisions. This disruption in the human–rover control loop reduces navigational accuracy, operational efficiency, and overall mission safety.

The primary contributors to this issue include the physical limits of signal transmission over large distances, teleoperation systems that are not designed for delayed feedback, and variability within network infrastructure such as latency fluctuations, packet loss, and jitter. These factors introduce uncertainty and desynchronization between command execution and feedback reception. Additionally, human operators experience reduced situational awareness due to delayed information, further increasing the likelihood of control errors.

The impact of these challenges is significant. Communication latency limits the operator's ability to respond to environmental hazards such as uneven terrain, steep slopes, and craters, increasing the risk of rover damage or mission failure. These risks are amplified by the harsh lunar environment and restricted operational windows, which place strict constraints on mission timelines and system reliability.

To address this problem, the integration of an AI-driven digital twin is proposed. This system creates a predictive virtual representation of the rover, enabling real-time estimation of its current state despite communication delays. By incorporating RGB-D sensor data and machine learning models, the digital twin enhances environmental awareness and supports more accurate decision-making. Ultimately, this approach aims to mitigate the operational impact of latency, improving rover safety, efficiency, and mission success while establishing a foundation for future deep-space exploration systems.

2.2 Goal Statement

The goal of this project is to improve the operational performance of a remotely operated RASSOR rover system by addressing the disruption in the human-rover control loop caused by Earth-to-Moon communication latency. As described in sections 2.1-2.4, latency forces operators to make decisions based on outdated state information, which increases navigation error, destabilizes control, and reduces mission reliability.

To tackle in-field communication latency, this project integrates a predictive digital twin capable of estimating the rover's current state during communication delay. The digital twin will be informed by real-time environmental and positional data captured through an OAK-D Lite RGB-D sensing system. This sensor enables synchronized color and depth data acquisition, allowing the system to collect spatial mapping information and rover motion data necessary to train and refine the predictive model.

By analyzing the teleoperation process and incorporating sensor-driven state prediction, the project aims to improve navigation accuracy, enhance operator situational awareness, and increase overall mission performance under simulated lunar latency conditions. The anticipated organizational benefit for the Florida Space Institute (FSI) is improved rover safety, reduced operational risk, and enhanced reliability of remote robotic exploration systems operating in latency-constrained environments.

2.3 Project Objectives (Unchanged)

- To demonstrate reliable rover operation using manual-only rover operation and a predictive digital twin system under approximately 3-second simulated Earth-to-Moon communication latency by April 15th, 2026
- To reduce obstacle collision frequency, as measured through controlled trial comparisons under latency conditions, between manual-only rover operation, digital twin-assisted operation, and AI-implemented digital twin operation by April 15th, 2026
- To reduce the average obstacle course completion time under latency conditions between the manual-only rover operation, digital-twin assisted operation, and AI-implemented digital twin operation by April 15th, 2026
- To reduce average path deviation from the predefined optimal route under latency conditions using digital-twin assisted operation, compared to manual-only rover operation, and AI-implemented digital twin operation by April 15th, 2026

2.4 Project Scope

Throughout the course of this project, the team will

- Construct one functional RASSOR rover
- Setup an internet based teleoperation control of the rover
- Establish Artificial communication delays in the rover communication
- Integrate an RGB-D sensing system
- Develop a digital twin framework to estimate rover state during communication gaps
- Collect and analyze performance metrics
- Document system architecture, testing procedures, and results
- Provide recommendations for continuation of the project in the second semester

2.5 Outside Project Scope

The following items are explicitly excluded from this project

- Develop systems in actual lunar or space environments
- Utilize space qualified hardware and systems
- Fabricate custom hardware outside the rover platform and sensor mounts
- Conducting durability testing on the rover
- Development of satellite communication infrastructure
- Full autonomous replacement of human operated control systems

2.6 Project Assumptions

- Internet Connectivity is sufficiently stable for teleoperation experiments
- Operators will have a sufficient baseline skill prior to testing rover operation
- Indoor or controlled outdoor test environments reasonably approximate key rover navigation challenges
- RGB-D sensors provide sufficient accuracy for obstacle detection and mapping
- An artificial approximate three second delay adequately represents lunar delay for testing purposes

2.7 Engineering Design Considerations

The system design is guided by key engineering requirements and constraints:

Functional Requirements

- Real time acquisition of RGB-D sensor data
- Accurate estimation of rover state during communication delays

- Implementation of controllable latency within the communication system
- User interface for effective remote teleoperation

Non-Functional Requirements

- Reliability under delayed communication conditions
- Usability for human operators
- Robustness to network variability and data inconsistencies

Risks and Limitations

- Prediction inaccuracies within the digital twin model
- Potential operator overreliance on predicted states
- Sensor calibration errors affecting mapping and estimation accuracy

3. DMADV - Measure: The Current State of the Process

Utilizing Issac Sim to run simulations of potential courses and obstacles to train the AI to maneuver by itself. The data being collected will be through the delays in manually moving the rover to simulate the time delay from Mexico, the amount of deviation from planned paths, and number of collisions with obstacles.

3.1 Current Performance Level

In previous simulations for the RASSOR rover design, the delay was only for 3 second delay. This project aims to have a baseline of 3 second delay, but could increase that time. The will be conducting a baseline latency testing with the metric being the time between the user input and the rover's movement. The team will be conducting a sample size of 50 trials to see significance in the latency. The collection period will be conducted for 3 days of controlled testing. The methodology used will be timestamp logging on the controller and rover. The project will handle minor timestamp drift and missing data. The timestamp drift will be corrected through synchronization. The latency will be shown through a histogram to showcase the distribution of the time delay.

The deviation from planned paths will have a metric of the percent of deviation from the rover's intended path to avoid obstacles. The sample size will be 25 autonomous navigation trials where the rover will go through an assortment of obstacles and try to reach the end. The collection period will be through 4 field sessions. The methodology used will be a planned path made by predefined waypoints, the rover's path being recorded through positional tracking, and deviation computed through the equation, (actual path error)/(planned path length) * 100%. The project will combat occasional loss of tracking while the rover is doing any sharp turns. The deviation will be shown through a pareto chart to determine the key causes of why the rover collided and see how the team can minimize some possible causes such as sensor misclassification or motor overshoot.

The rover will need to be able to reduce the number of collisions. The metric will be the count of collisions with obstacles in both manual and autonomous operations. The sample size will be 30 total trials with 10 being autonomous trials and 20 being manual trials through a collection period of 3 days. The methodology will be through manual observation. The project will try to handle false positives of potential obstacles and will need to have all collisions verified to ensure accuracy. The collision number will be shown through a run chart of collision frequency over time. The count could potentially fluctuate and needs to be determined if it is a chronic issue.

3.2 Key Variables

There are multiple key variables needed to ensure that the rover is functioning properly when driven with the and without the digital twin and improve on the navigation even under time delay conditions. The key performance output variables (KPOVs) will be the rover response latency, deviation from the planned paths, and the amount of collisions with obstacles. The Key performance input variables (KPIVs) are the sensor processing time, motor calibration, AI inference timing, and obstacle detection threshold.

3.3 Target Performance Levels

- The latency even with more than a 3 second delay does not impact the rover’s motor function and is able to avoid any obstacles.
- Deviation from the planned path is minimal with a less than 5% deviation.
- The number of collisions is at 0.

4. DMADV - Analyze: The Current State of the Process

At this stage, the Analyze phase focuses on finding the input variables that most affect rover performance under simulated Earth-to-Moon communication delay. It also evaluates which design concept can best meet project needs. Since the physical rover is still being built, this analysis relies on the defined system requirements, early virtual testing in Isaac Sim, and the baseline performance measures established in the report.

The main Key Performance Input Variables (KPIVs) are communication delay, rover speed, obstacle density, sensor accuracy, and digital twin prediction accuracy. The main Key Performance Output Variables (KPOVs) are collision frequency, task completion time, and path deviation from the intended route. Among these inputs, communication delay is the most critical because it forces the operator to make decisions based on outdated feedback. As the delay increases, the rover travels farther before any corrections can be made, which raises navigation error, leads to repeated corrective actions, and increases the risk of collisions. These issues become worse when the rover moves faster or when there are more obstacles in the environment. Additionally, inaccurate RGB-D sensing or poor digital twin predictions weaken the reliability of the rover's state estimate, which further lowers the operator's awareness and control.

This analysis shows that the main performance gap in the current manual-only design is the difference between the rover’s actual state and the operator’s perceived state during delayed teleoperation. From an engineering viewpoint, latency weakens the control loop that includes the human operator and reduces navigation stability. From a data science standpoint, raw camera feedback alone is not enough; the system also needs state estimation and short-term predictions using sensor and motion data. Therefore, preliminary simulation-based analysis indicates that improving performance relies less on changing rover mobility and more on reducing uncertainty in the rover's position and future movement during the communication delay.

Three design directions were considered: an improved operator feedback interface, AI-assisted path planning, and a predictive digital twin. The improved interface enhances visibility but still mainly depends on delayed feedback. AI-assisted path planning could lessen the operator's workload, but it adds more complexity to implementation and relies heavily on the model's robustness across different environments. The predictive digital twin stands out as the best option because it directly tackles the main performance issue, loss of situational awareness caused by latency, while keeping the human operator involved. For this reason, the preferred design direction is a predictive digital twin supported by RGB-D sensing and synchronized rover state data. The key design priorities moving forward are prediction accuracy, reliability of sensor data, and synchronization between the physical rover and its virtual model.

5. *DMADV - Design: Preliminary Process Design Alternatives*

Design Requirements Derived from Measure and Analyze

Based on the Measure and Analyze phases, the primary limitation is the loss of situational awareness caused by communication latency. The six-second round-trip delay disrupts the human–rover control loop, increasing path deviation, collision frequency, and task completion time.

Key performance outputs (KPOVs) include latency impact, path deviation, and collision frequency, while key inputs (KPIVs) include communication delay, sensor accuracy, rover speed, and prediction accuracy. Among these, communication delay and uncertainty in rover state were identified as the main drivers of performance degradation.

To address this, the system must reduce uncertainty in rover position and future movement, improve operator awareness under delay, and integrate real-time RGB-D data from the OAK-D Lite sensor into the control system.

Current RASSOR Rover Design

The baseline system reflects traditional rover operation, where the operator manually controls the rover using delayed visual feedback from onboard cameras. As shown in Figure 1, commands and feedback each experience a three-second delay, resulting in outdated system awareness.

This approach leads to increased navigation error, higher collision frequency, and inefficient operation due to repeated corrective actions. While recent work has introduced simulation environments and sensor integration, the control strategy remains reactive and dependent on delayed feedback. This baseline establishes the reference point for evaluating improved system designs.

Real-Time Predictive Digital Twin System

This design builds directly on the current system by integrating a predictive digital twin using real-time RGB-D data from the OAK-D Lite camera and rover motion data. The digital twin, developed within Isaac Sim, creates a synchronized virtual representation of the rover and its environment and predicts future rover states during communication delay.

This approach directly addresses the key performance gap identified in the Analyze phase: the mismatch between the rover's actual state and the operator's perceived state. By providing predictive visualization, the operator can make decisions based on estimated current conditions rather than delayed feedback, improving navigation accuracy and reducing collisions.

This design aligns with the current phase of the project, where simulation infrastructure and sensor data pipelines are already being developed. However, it requires accurate synchronization between physical and virtual systems, reliable sensor data, and sufficient computational capability to maintain real-time prediction.

AI-Enhanced Digital Twin with Assisted Navigation

This design extends the digital twin framework by incorporating AI-driven navigation. In this system, the digital twin not only predicts rover state but also assists in decision-making by generating optimized paths using environmental data from the RGB-D sensor.

The operator transitions from direct control to supervisory control, providing high-level commands while the system executes navigation tasks autonomously. This reduces the impact of communication delay by minimizing reliance on continuous manual input.

This approach has the potential to further reduce path deviation and collision frequency beyond digital twin-assisted operation. However, it introduces increased system complexity and depends on the accuracy and robustness of machine learning models. As such, it is considered a future phase built upon the successful implementation of the digital twin framework.

Enhanced Manual Teleoperation with Sensor-Augmented Feedback

This design improves the baseline manual system by incorporating RGB-D sensor data into the operator interface without predictive modeling. The OAK-D Lite camera provides spatial mapping and obstacle detection, which can be visualized to improve operator awareness.

This approach enhances perception compared to traditional camera-only feedback and may reduce navigation errors. However, it does not resolve the core issue of delayed feedback, as the operator still relies on outdated information when making decisions.

While this design is the most feasible and easiest to implement, it provides only incremental improvements and does not fully address the performance limitations identified in the Analyze phase.

Trade-off, Selection, and High-Level Design Direction

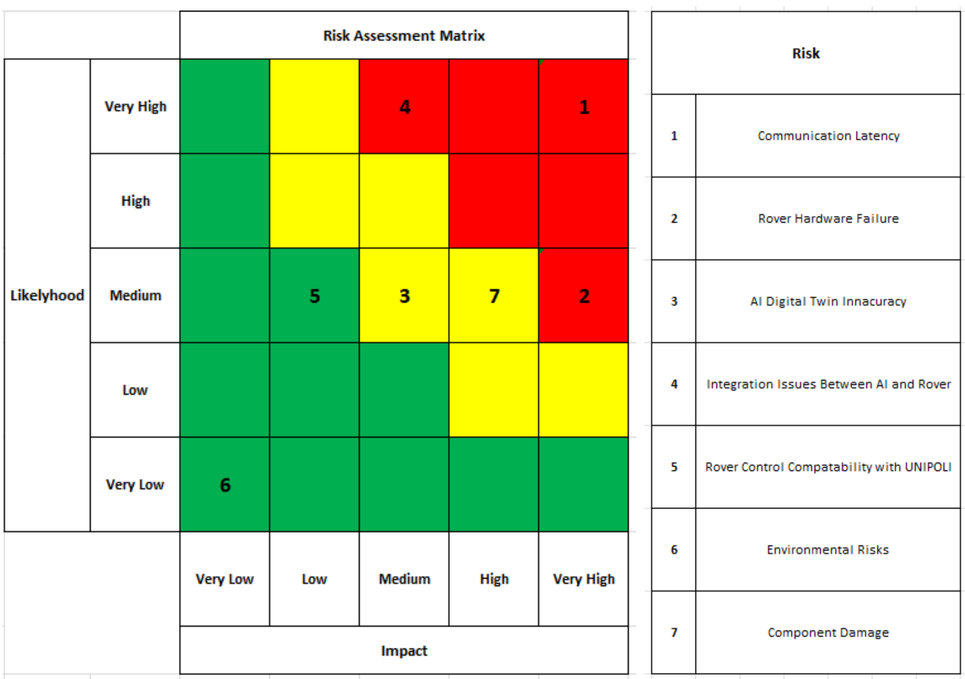
The three design alternatives represent increasing levels of system capability. The enhanced manual system improves perception but does not eliminate latency-related issues. The AI-enhanced digital twin offers the highest level of autonomy and performance improvement but introduces significant complexity and depends on future development.

The predictive digital twin-assisted system provides the most balanced solution by directly addressing the root cause of performance degradation while remaining feasible within the current project scope. It reduces uncertainty in rover state, improves operator awareness, and integrates with the existing simulation and sensor infrastructure.

Based on this evaluation, the predictive digital twin-assisted system is selected as the preferred design. The high-level system will integrate real-time RGB-D data and rover state information into a synchronized digital twin within Isaac Sim, enabling predictive feedback under communication delay conditions.

Future work will focus on improving prediction accuracy, ensuring system synchronization, and validating performance through controlled testing. This framework will also support the transition to AI-assisted navigation in subsequent project phases.

6. Risk Management (Project)



		Risk Assessment Matrix				
Likelyhood	Very High			4		1
	High					
	Medium		5	3	7	2
	Low					
	Very Low	6				
		Very Low	Low	Medium	High	Very High
		Impact				

Risk	
1	Communication Latency
2	Rover Hardware Failure
3	AI Digital Twin Innacuracy
4	Integration Issues Between AI and Rover
5	Rover Control Compatability with UNIPOLI
6	Environmental Risks
7	Component Damage

Green - Low Risk
Yellow - Possible Work Around
Red - High Risk

Figure 2: Risk Assessment Matrix

The Primary risks to the project relate to communication latency, hardware reliability, AI accuracy, Integration, Cross functional coordination and environmental factors. Risk levels are determined based on Likelihood and impact. Mitigation strategies are implemented to reduce the probability of occurrence or the severity of the consequences.

Communication Latency is a primary risk resulting from the simulated delay conditions. There is a high likelihood that operators will experience delayed feedback, which may reduce real-time responsiveness and increase the risk of incorrect maneuvering. The impact of this risk is high because it directly affects rover control and mission performance. To mitigate this risk, the team will implement AI-assisted prediction through a digital twin model that simulates rover motion ahead of the physical execution. This will allow the team to be proactive when making decisions despite the latency.

Rover Hardware failure can be a significant risk because of potential network instability or hardware malfunction during teleoperation. The likelihood of this is moderate but the impact is high because hardware failure could slow testing, damage equipment, or interrupt the presentation from the different universities. This risk might come from communication interruptions, electrical issues or mechanical component stress. To mitigate this risk, a rover failsafe safe stop mode will be implemented to immediately pause motion and stabilize the system if a fault or communication loss is detected.

The risks associated with delays in camera delivery and 3D printing are now absent from the project, as these tasks have been completed. The printing of the rover has been completed, and all parts have been delivered on time. In their place, the team now has the risk that critical rover components may become damaged during testing or operation due to mechanical stress, handling, or environmental conditions. The likelihood of this risk is moderate, and the impact is medium to high, as damaged components may require replacement, leading to delays in testing and potential schedule disruption. To reduce this risk, the main body of the rover is designed for modular replacement, so it can be easily replaced through new printed parts. Specific pieces will be printed to shield vital technical components, such as the camera system, to protect them from damage. Additionally, careful handling procedures are implemented to minimize the likelihood of damage.

AI digital twin inaccuracy is a moderate risk because of limited data training during the early phases. If the model inaccurately predicts rover behavior or terrain interaction, operator decision making could be negatively affected. The impact is medium because inaccuracies may reduce system reliability but are unlikely to cause major failure due to human supervision. Mitigation includes maintaining human oversight and continuously refining the AI model with real sensor data to improve prediction accuracy.

Integration issues between AI and rover can present a moderate risk with medium impact. Software compatibility issues, data formatting inconsistencies, or control interface mismatches may show up during system integration. These issues could delay testing and reduce overall system efficiency. To mitigate this risk, the team will conduct incremental testing throughout development, validating individual subsystems before full integration.

Because this project involves cross-institution collaboration, differences in rover configuration may create compatibility issues. The probability of this is moderate, and the impact is medium since this could delay joint teleoperation demonstrations. This risk comes from configuration differences and coordination challenges. Mitigation will involve maintaining open and consistent communication between teams and potentially joint testing sessions. Environmental risks such as unfavorable weather conditions affecting outdoor testing, are considered low likelihood and low impact due to the availability of controlled indoor facilities. Testing conducted in the Exolith lab environment reduces exposure to environmental variability so this means the mitigation for the risk is to prioritize indoor testing and scheduling flexibility.

7. Summary

This project tackles the challenge of operating a lunar rover remotely while dealing with communication delays between Earth and the Moon. Delayed feedback lowers the operator's awareness of the situation, which raises the chances of navigation mistakes, collisions with obstacles, and inefficient task completion. To solve this problem, the team proposed a predictive digital twin that uses RGB-D sensors within a RASSOR rover system. The aim is to improve navigation accuracy, reduce collisions, and decrease path deviation under simulated latency conditions.

To set a baseline, the team defined the process, project scope, requirements, and constraints. They then developed a measurement plan using Isaac Sim, which involved simulating delays, logging timestamps, and comparing manual-only, digital twin-assisted, and AI-assisted rover operations. From this baseline analysis, the team identified communication delay as the main factor affecting performance. Rover speed, obstacle density, sensor quality, synchronization, and prediction accuracy were significant secondary factors. The findings indicated that the main issue isn't the rover’s mobility but the gap between its actual state and the operator’s perception during delayed control.

After considering different options, the team selected the predictive digital twin as the best design because it directly addresses the loss of situational awareness caused by latency while keeping human control in the loop. Although final testing is not yet complete, the implementation is expected to lower collision rates, enhance path tracking, and decrease task completion times compared to manual-only operation. These improvements should boost rover safety, operator effectiveness, and the overall reliability of the mission.

The project's main strengths are its use of the DMADV framework, its combination of engineering design with data-based analysis, and its assessment of various design alternatives. Its current limitations are that the physical rover is still being developed and much of the existing evidence relies on simulations rather than completed experimental trials. The final analysis will focus on finishing system integration and validating performance through controlled testing under simulated delays. Some open questions include prediction errors in the digital twin, subsystem synchronization, hardware reliability, and endurance in different obstacle environments.

7.1 Standards

The fabrication of rover components using additive manufacturing will adhere to 3D printing safety guidance provided by Stanford Environmental Health & Safety. These guidelines identify potential hazards associated with 3D printing, including emissions of ultrafine particles, volatile organic compounds, and thermal risks from heated components. To mitigate these risks, printing will be conducted in a well-ventilated environment with appropriate operational controls and adherence to manufacturer specifications. Implementing these measures ensures safe production of prototype components while minimizing health and environmental risks.

ACM Code Of Ethics

The proposed system adheres to the ACM Code of Ethics and Professional Conduct for responsible computing practice. The code emphasizes maintaining high standards of professional competence, ethical behavior, and integrity, as well as respecting the intellectual work required to produce new ideas, inventions, creative works, and computing artifacts. Incorporating these principles ensures that the rover framework is developed and implemented in a manner that is responsible, reliable, and consistent with accepted professional and ethical standards.

NASA Material Safety Data Sheet

Testing in simulated lunar soil will follow the NASA Material Safety Data Sheet (MSDS) for JSC-1A lunar regolith simulant used in the Exolith Lab. The guidelines recommend minimizing dust inhalation, washing exposed skin after handling, and using appropriate personal protective equipment to ensure safe operation during rover testing in a simulated lunar environment.

7.2 Recent Topics and Trends

One of the most significant trends in engineering today is the use of digital twin technology to enhance the performance, flexibility, and decision-making of complex systems. A digital twin is essentially a virtual replica of a physical system that continuously integrates real-world data to simulate, predict, and optimize its behavior (Xu et al., 2023). This technology allows engineers to monitor and analyze systems in real time without disrupting their physical operation, making it especially valuable for robotics and industrial applications.

In manufacturing and robotics, digital twins are increasingly used to connect physical machines with their virtual counterparts. Xu et al. (2023) describe how high-fidelity virtual models linked to real-time sensory data can improve control and optimization of robotic systems, enhancing both responsiveness and reliability. This capability is directly relevant to Industrial Engineering, where managing complex automated systems efficiently is a core goal. Digital twins allow engineers to test design alternatives virtually, simulate “what-if” scenarios, and make data-driven decisions that reduce errors and downtime.

Beyond robotics, digital twin technology is a cornerstone of modern cyber-physical systems and Industry 4.0 initiatives. Advanced digital twins incorporate artificial intelligence, machine learning, and IoT data to monitor performance, predict failures, and optimize operations across the system lifecycle (Digital twin technology advancing industry 4.0 and industry 5.0, 2025). This aligns closely with Industrial Engineering objectives, such as improving system reliability, optimizing processes, and integrating data from multiple sources for better operational decision-making.

Recent research also highlights the growing role of AI-enhanced digital twins for adaptive control and predictive management in complex systems (Dong et al., 2023). By combining real-time sensing with virtual simulation, these systems can anticipate and respond to dynamic changes, supporting more resilient and efficient operations. For Industrial Engineering, this provides powerful tools to analyze performance, improve process efficiency, and ensure system robustness in environments where uncertainty and variability are common.

Overall, digital twin technology represents a major advancement for Industrial Engineering. It transforms traditional approaches to process design, quality control, and system optimization into a predictive, data-driven framework. As industrial and robotic systems become more interconnected and sophisticated, digital twins offer engineers a way to better understand system behavior, optimize performance, and improve decision-making in real time (Xu et al., 2023; Digital twin technology advancing industry 4.0 and industry 5.0, 2025; Dong et al., 2023).

8. Broader Impacts on Sustainability and Beyond

Safety Factors

Predictive modeling plays a critical role in improving rover safety and reducing the risk of mission failure. Lunar surface environments present significant hazards such as craters, steep slopes, uneven terrain, and regions without solar illumination. These conditions increase the likelihood of rendering a rover inoperable and compromising mission objectives. Communication latency between Earth and the Moon introduces safety risks by creating delays in command execution and sensor feedback. This desynchronization can lead to inaccurate control decisions, increasing the probability of unsafe operations.

The implementation of a predictive safety system such as an AI-driven digital twin helps mitigate these risks by estimating the rover's future state during communication delays. This predictive capability allows operators to anticipate potential hazards before they occur, improving situational awareness and enabling more informed decision-making. As a result, the system enhances operational safety, improves mission reliability, and supports long term sustainability of rover operations

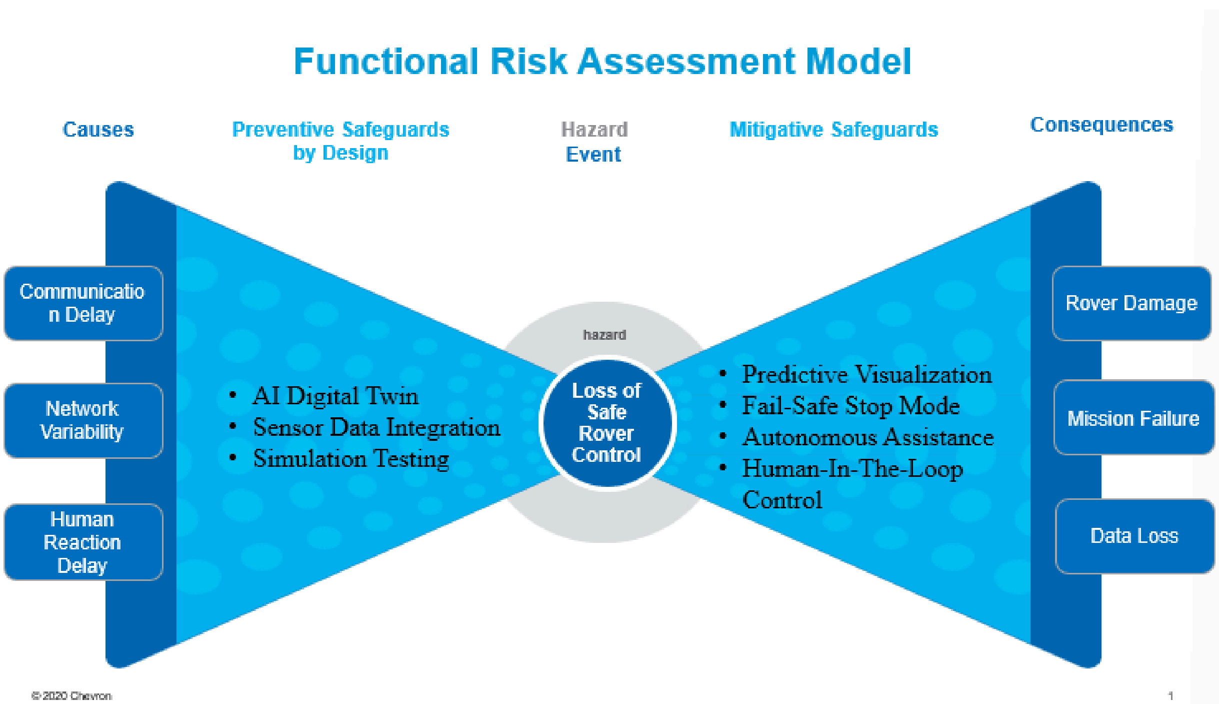


Figure 3: Bowtie Functional Risk Assessment Model

Public Health Factors

Reducing human exposure to hazardous environments through the use of remote robotics supports the long-term sustainability of human space exploration. Improved rover autonomy enables more efficient support of human life by reducing the need for astronauts to perform high-risk tasks in extreme conditions. Such as exposure to radiation and unpredictable environments. This would ultimately reduce the physical strain and minimize health risks for the crews that would oversee these missions.

On the other hand, the advancements in robotics developed for space exploration also have significant implications for public health on Earth. The robotic technologies designed for remote operation and hazard mitigation can be adapted for use in disaster response and medical applications. These robotics could reduce the need for human intervention in dangerous events. By enhancing the reliability and autonomy of robotic systems, this supports broader public health goals by minimizing human exposure and contributing to safer working conditions both in space and on Earth.

Ethics and Welfare Factors

Adherence to the NASA ethics framework is essential for ensuring responsible and ethical system design. This framework emphasizes the principles of beneficence, respect for people, and justice, requiring that the engineering decisions maximize overall benefit, respect individual rights, and remain consistent with established standards of fairness. If these principles are not upheld the system must be reevaluated to ensure ethical compliance.

In the context of autonomous remote rover operations with artificial intelligence (AI), maintaining a human-in-the-loop approach is critical. The purpose of integrating AI is not to replace human operators but to enhance decision-making through predictive insights and improved situational awareness. This ensures that human judgment remains at the forefront while leveraging AI to improve accuracy and efficiency. Responsible system design requires transparency, reliability, and accountability in how AI models are developed and applied. By supporting rather than replacing human operators, the proposed digital twin framework aligns with ethical engineering practices and promotes the safe and responsible use of AI. This approach contributes to long-term sustainability of space exploration by ensuring that these advancements align with human values and societal well-being.

Global Factors

Space exploration is inherently a global effort, and effective solutions must be scalable internationally. Open communication and collaboration between nations are essential for improving space exploration efficiency, reducing mission costs, and advancing robotic technologies that can be applied to challenges on Earth. International partnerships enable the sharing of knowledge, resources, and technical expertise leading to increased productivity and extended mission lifespans. The development of a lunar research base highlights the importance of global cooperation in advancing human exploration. Technologies such as In-Situ Resource Utilization (ISRU), which involve the harvesting and processing of local resources such as water, ice, regolith on the Moon, Mars, and asteroids are critical for establishing a sustainable human presence in deep space. These efforts require coordinated international collaboration to ensure scalability, efficiency, and long-term success.

Cultural Factors

Cross-cultural collaboration enhances innovation and problem-solving by bringing together diverse perspectives and approaches to complex engineering challenges. Diverse cultural backgrounds promote inclusivity, broaden awareness, and foster an environment that encourages creative thinking and technological advancement. International collaboration in engineering requires coordination across different methodologies, communication styles, and problem-solving strategies. These differences when effectively managed, strengthen teamwork and lead to more resilient system designs. By integrating multiple perspectives, teams are better equipped to address uncertainties and develop stronger solutions. In the context of space exploration, cross-cultural collaboration supports the long-term sustainability of global missions by enabling the exchange of knowledge and innovation across nations. This collaborative approach not only enhances the development of advanced technologies but also reinforces a shared commitment to advancing humanity’s presence beyond Earth.

Social Factors

Reliable robotic systems enhance human safety while expanding scientific capabilities in space exploration. These systems form the foundation for the development of increasingly autonomous and self-reliant rovers by enabling deeper space exploration without requiring the presence of humans in hazardous conditions. As robotic exploration advances, it reduces the need for astronauts to be exposed to extreme conditions such as radiation and unknown terrain. In turn this supports safer and more sustainable mission operations over time.

High-profile rover/space missions are a great way to capture the public eye by inspiring future generations of scientists and engineers. Tangible human achievements would motivate students to pursue careers in science, technology, engineering, and mathematics (STEM). The visibility of technological achievements in space fosters educational interest and promotes innovation across disciplines.

Beyond the interpersonal social factors, space exploration has been pivotal in driving sustainable practices on Earth. For example, space-based solar power systems have the potential to capture solar energy without atmosphere interference and transmit clean energy back to Earth, supporting long-term energy sustainability while minimizing reliance on non-renewable resources” (Sierra Space, 2024). The continued development of robotics and artificial intelligence for space applications not only advances scientific discovery but also drives technological innovations with broader societal benefits.

Environmental Factors

The establishment of a lunar base is critical for enabling efficient resource extraction while reducing the environmental impact associated with repeated space missions. Through the establishment of a lunar base, future missions can reduce dependence on Earth-based launches, therefore minimizing the environmental footprint of space exploration.

Transporting materials from Earth into orbit and beyond contributes significantly to environmental degradation. Results from a study indicate that rocket launches impact the environment through increased carbon emissions, pollution, and resource consumption. The environmental impact becomes more significant at higher orbital levels, with geostationary launches exhibiting the greatest ecological footprint compared to low Earth orbit missions (Donou-Adonsou, 2024). These impacts include increased greenhouse gas emissions, reduced biocapacity, and long-term strain on natural resources.

The development of lunar-based manufacturing supports environmental sustainability by reducing the need to transport materials from Earth. Furthermore, simulation and testing in controlled regolith environments improve mission success rates, minimizing failed missions that would otherwise contribute to unnecessary resource consumption and environmental harm. Ultimately supporting more sustainable space exploration practices by optimizing resource usage will limit environmental impact both on Earth and in space.

Economic Factors

Improving rover performance reduces mission costs and contributes to the development of a sustainable space economy. Investments in space exploration generate significant economic impact, creating a multiplier effect in which advancements in aerospace, robotics, and data science support job growth across multiple industries.

Transporting materials from Earth to the Moon and beyond is extremely costly. By establishing a lunar base capable of extracting resources such as water, oxygen, metals are essential for reducing mission expenses in turn facilitating long-term exploration. Reliable and efficient rover systems play a critical role in this process on the lunar surface. By improving rover reliability under communication latency, the proposed digital twin system contributes to reducing costly mission errors and enhancing overall operational efficiency.

Advancements in autonomous navigation further enhance economic efficiency by reducing the need for continuous ground control. Increasing autonomy improves mission success rates given the high cost of space missions. Initiatives such as NASA's Moon to Mars campaign highlight the importance of developing sustainable infrastructure and technologies on the moon to support future deep-space exploration

The Moon to Mars campaign is a long-term initiative to establish a sustainable human presence on the Moon through the Artemis program to prepare and pioneer the first human missions to Mars. Through lunar exploration to test operational capabilities, infrastructure, and technologies that will be vital for deep space travel. NASA first ever agency-wide economic impact report indicates that the space agency generated more than $64.3 billion in total economic output during the 2019 fiscal year, supporting more than 312,000 jobs nationwide, and generated an estimated $7 billion in federal, state, and local taxes throughout the United States" (ResearchFDI et al., 2025). Continued advancements will only further expand these economic benefits and contribute to a sustainable long-term space economy.

9. Project Management and Status of the Design Project

9.1 Project Timeline

The project schedule is organized using a detailed Gantt chart that outlines all major tasks, milestones, durations, dates, dependencies, and assigned responsibilities from January through April 2026. The schedule reflects a structured and milestone-driven approach to ensure the successful development of the RASSOR rover system and its associated digital twin framework within the academic semester.

As of the current reporting period, the project is approximately halfway complete and is progressing on schedule with respect to planned milestones. Early project phases, including project initiation, proposal development, budgeting, and client presentations, have been completed at 100%. These completed tasks established a strong foundation for both the technical and analytical components of the project.

The team is currently focused on rover hardware development, specifically finalizing the base structure and integrating electronic components. Mechanical assembly of the base and wheels is approximately 90% complete, while implementation of electronic components is approximately 80% complete. Current efforts include integration of onboard processing hardware, such as the Le Potato microcontroller, and ensuring compatibility between hardware and control systems.

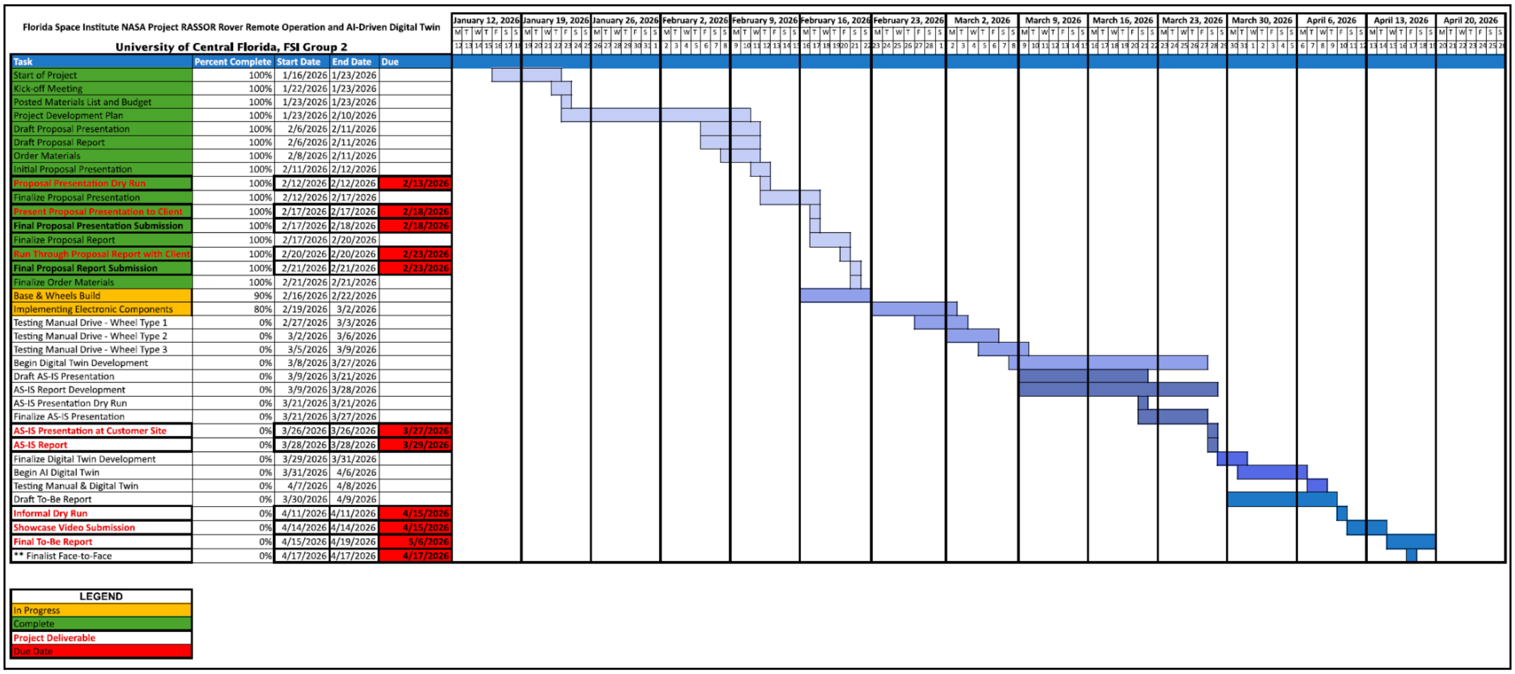
Some tasks, particularly those related to rover hardware development, have experienced minor delays due to external factors such as extended 3D printing times and temporary shortages of materials, including printing filament. These delays primarily affected the timeline for mechanical assembly and initial testing. However, the team has proactively adjusted the project schedule by reallocating time and resources across parallel tasks, allowing progress in other areas such as planning, documentation, and preliminary system integration to continue. As a result, the overall project timeline remains on track, and the team anticipates meeting all major milestones and final deliverables within the planned schedule.

The next phase of the project involves experimental testing and data collection, including comparative testing of multiple wheel designs under simulated operating conditions. These tests will evaluate performance metrics such as mobility, stability, and efficiency, providing critical data for the Measure and Analyze phases of the project. In parallel, development of the digital twin system is scheduled to begin, with the goal of creating a predictive model capable of simulating rover behavior under communication delay conditions.

In addition to technical development, the project includes structured reporting and presentation milestones. Key upcoming deliverables include the AS-IS Presentation and Report in late March, followed by the To-Be Report, Showcase Video, and Finalist Face-to-Face presentation in April. These milestones ensure continuous documentation, evaluation, and communication of project progress to stakeholders.

The project schedule demonstrates coordinated parallel workstreams, where mechanical, electrical, software, and analytical tasks are developed concurrently. This approach enables efficient use of time and resources while supporting integration across system components. Task sequencing has been carefully planned to align dependencies, ensuring that testing, modeling, and analysis phases build upon completed hardware and data collection efforts.

The project remains on track to meet its planned deadlines. The Gantt chart reflects a realistic and well-structured execution plan designed to deliver a fully functional rover prototype, a validated digital twin framework, and comprehensive analysis and documentation of system performance. Continued adherence to the schedule, along with effective team coordination and resource management, is expected to support successful project completion.



Task	Percent Complete	Start Date	End Date	Due
Start of Project	100%	1/16/2026	1/23/2026	
Kick-off Meeting	100%	1/22/2026	1/23/2026	
Posted Materials List and Budget	100%	1/23/2026	1/23/2026	
Project Development Plan	100%	1/23/2026	2/10/2026	
Draft Proposal Presentation	100%	2/6/2026	2/11/2026	
Draft Proposal Report	100%	2/6/2026	2/11/2026	
Order Materials	100%	2/8/2026	2/11/2026	
Initial Proposal Presentation	100%	2/11/2026	2/12/2026	
Proposal Presentation Dry Run	100%	2/12/2026	2/12/2026	2/13/2026
Finalize Proposal Presentation	100%	2/12/2026	2/17/2026	
Present Proposal Presentation to Client	100%	2/17/2026	2/17/2026	2/18/2026
Final Proposal Presentation Submission	100%	2/17/2026	2/18/2026	2/18/2026
Finalize Proposal Report	100%	2/17/2026	2/20/2026	
Run Through Proposal Report with Client	100%	2/20/2026	2/20/2026	2/23/2026
Final Proposal Report Submission	100%	2/21/2026	2/21/2026	2/23/2026
Finalize Order Materials	100%	2/21/2026	2/21/2026	
Base & Wheels Build	90%	2/16/2026	2/22/2026	
Implementing Electronic Components	80%	2/19/2026	3/2/2026	
Testing Manual Drive - Wheel Type 1	0%	2/27/2026	3/3/2026	
Testing Manual Drive - Wheel Type 2	0%	3/2/2026	3/6/2026	
Testing Manual Drive - Wheel Type 3	0%	3/5/2026	3/9/2026	
Begin Digital Twin Development	0%	3/8/2026	3/27/2026	
Draft AS-IS Presentation	0%	3/9/2026	3/21/2026	
AS-IS Report Development	0%	3/9/2026	3/28/2026	
AS-IS Presentation Dry Run	0%	3/21/2026	3/21/2026	
Finalize AS-IS Presentation	0%	3/21/2026	3/27/2026	
AS-IS Presentation at Customer Site	0%	3/26/2026	3/26/2026	3/27/2026
AS-IS Report	0%	3/28/2026	3/28/2026	3/29/2026
Finalize Digital Twin Development	0%	3/29/2026	3/31/2026	
Begin AI Digital Twin	0%	3/31/2026	4/6/2026	
Testing Manual & Digital Twin	0%	4/7/2026	4/8/2026	
Draft To-Be Report	0%	3/30/2026	4/9/2026	
Informal Dry Run	0%	4/11/2026	4/11/2026	4/15/2026
Showcase Video Submission	0%	4/14/2026	4/14/2026	4/15/2026
Final To-Be Report	0%	4/15/2026	4/19/2026	5/6/2026
** Finalist Face-to-Face	0%	4/17/2026	4/17/2026	4/17/2026

LEGEND
In Progress
Complete
Project Deliverable
Due Date

9.2 Project Management Plan

An Integrated Master Schedule (IMS) is attached to this report for reference. The schedule provides a detailed breakdown of the project Gantt chart, including itemized tasks, subtasks, defined milestones, and assigned responsibilities for each activity. This structured schedule ensures clear task ownership, accountability, and alignment across hardware development, digital twin implementation, and reporting deliverables throughout the project lifecycle. The project is approximately 50% complete, with the rover build and electronic component implementation underway, while upcoming testing and digital twin development tasks remain on schedule.

10. Acknowledgments

The team gives their thanks to all persons who have helped in preparing this proposal, and in the formation of this project:
Dr. Luis Rabelo
Dr. Mansooreh Mollaghasemi
FSI Representative, Christian Vincent
Project Support, Carson Gray

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Appendices

Project Charter

Department of Industrial Engineering and Management Systems

College of Engineering and Computer Science

University of Central Florida

Industrial Engineering Capstone Senior Design/Praxis of Data Science Course

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SENIOR DESIGN PROJECT CHARTER

Project Sponsor:	*Florida Space Institute*

Project Manager:	*Jennifer Cifuentes*		
Co – Project Manager			
E-mail Address:	*jennifer.cifuentes@ucf.edu*		
Phone Number:	*+1 (321)900-2878*		
Process Impacted:	*Enter the name of the impacted process(es) here*		
Expected Start Date:	*Remote rover teleoperation and state monitoring under simulated lunar communication delay conditions*		
Expected Completion Date:	*April 19, 2026*		
Green Belts Assigned:	*Luis Rabelo, Ph.D.*	***Black Belts Assigned:***	*Ahmed Mohammed Ph.D.*
Mentor:	*Mansooreh Mollaghasemi, Ph.D.*	***Master Black Belt Assigned:***	*Jose Flores, Ph.D.*

Element	Description of Element	Project Charter
1. Title of Project	What is the title of the improvement project?	RASSOR Rover Remote Operation and AI-Driven Digital Twin Development
2. Project Description	Brief background and project's purpose	To establish methods of addressing time delay in lunar communications.
3. Process Description	Brief description of the process in which opportunity exists	Remote rover operation under simulated lunar conditions is impacted by communication latency and limited environmental feedback. Operators experience delays between issued commands and rover response, reducing efficiency and increasing operational risk. An opportunity exists to integrate a predictive digital twin

<table>
<tr><td></td><td></td><td>system to improve synchronization, situational awareness, and navigation performance.</td></tr>
<tr><td>4. Business Case</td><td>Brief explanation of the expected financial improvement, or other justification</td><td>Expected improvements include reduction of risk of rover damages, improvements to mission time duration, and efficiency of rover operation.</td></tr>
<tr><td>5. Benefit to External Customers</td><td>Who are the final customers and what benefits will they see?</td><td>FSI will receive a validated prototype system and documented methodology that can serve as a foundation for continued research in addressing communication latency.</td></tr>
<tr><td>6. Problem Statement</td><td>Problem and goal statement (project's purpose)</td><td>Current remote rover operations suffer from communication latency, limited environmental awareness, and lack of synchronized feedback systems, reducing operational efficiency and rover safety during remote navigation.

To tackle in-field communication latency by allowing the digital twin to predict the rover's current state, improving navigation, operator awareness, and overall mission performance.</td></tr>
<tr><td>7. ProjectObjectives</td><td>What are the performance metrics such as throughput, quality, inventory, defects, yield, costs, etc.? What improvement is targeted?</td><td>
<table>
<tr><th>Performance Metric</th><th>Current (As-Is) Value</th><th>Future (To-Be) Value</th></tr>
<tr><td>Operator Feedback During Latency</td><td>No predictive state feedback</td><td>Predictive digital twin visual</td></tr>
<tr><td>Positional Awareness</td><td>Lack of data collection</td><td>Integrated sensor-based state modeling</td></tr>
<tr><td>Navigation Efficiency</td><td>Manual drive only</td><td>Design predictive modeling architecture</td></tr>
<tr><td>System Integration</td><td>No autonomous</td><td>Integrated rover with digital twin system</td></tr>
</table>
</td></tr>
<tr><td>8. Scope Statement</td><td>Which parts of the process will be investigated and which parts will be excluded? What are the major tasks to complete?</td><td>The team will construct one functional RASSOR rover and set up for internet based teleoperation control of the rover.

The team will not develop systems in actual lunar or space environments or utilize space qualified hardware and systems</td></tr>
</table>

9. Proposed Methods	Which design and analysis concepts, techniques and tools are going to be used?	The team will use systems engineering principles, iterative prototyping, and structured testing to develop and validate a physics-based digital twin with steps towards AI-assisted state prediction. Performance testing and system integration will support evaluation.	
10. Key Deliverables	What are the tangible and/or intangible objects that are produced as a result of the project and that relate directly to and align with the objectives of the project?	A completed, drivable rover - The final rover built and operating, able to be driven, and outfitted with sensors for positional data acquisition. Digital Twin System - A physics based Digital Twin of the physical rover that can model the effects of motion commands on the rover. Architecture for AI assisted motion control - Setup for autonomous AI rover navigation, to be built upon in the second semester of the project. Handoff Documentation - Well organized documentation of the entire process of design and control, to allow the second semester to have the necessary information to take over the project.	
11. Schedule	Give the key project milestones and dates	Project Start Date:	1/16/26
	"D" – Define Phase	"D" Completion Date:	2/21/26
	"M" – Measure Phase	"M" Completion Date:	3/28/26
	"A" – Analyze Phase	"A" Completion Date:	3/31/26
	"D" – DesignPhase	"D" Completion Date:	4/8/26
	"V" – Verify Phase	"V" Completion Date:	4/11/26
12. Team Members	Names of the project team members (in alphabetical order of last names)?	Jennifer Cifuentes, Noah Choi, Sean Crites, Matteo D'Agostino, Mattias Escudero, Jonathan Gyure, Abdiel Joseph	

13. Costs and Resources	What are the estimated project costs and needed resources?	The estimated project cost will be $1461.55.

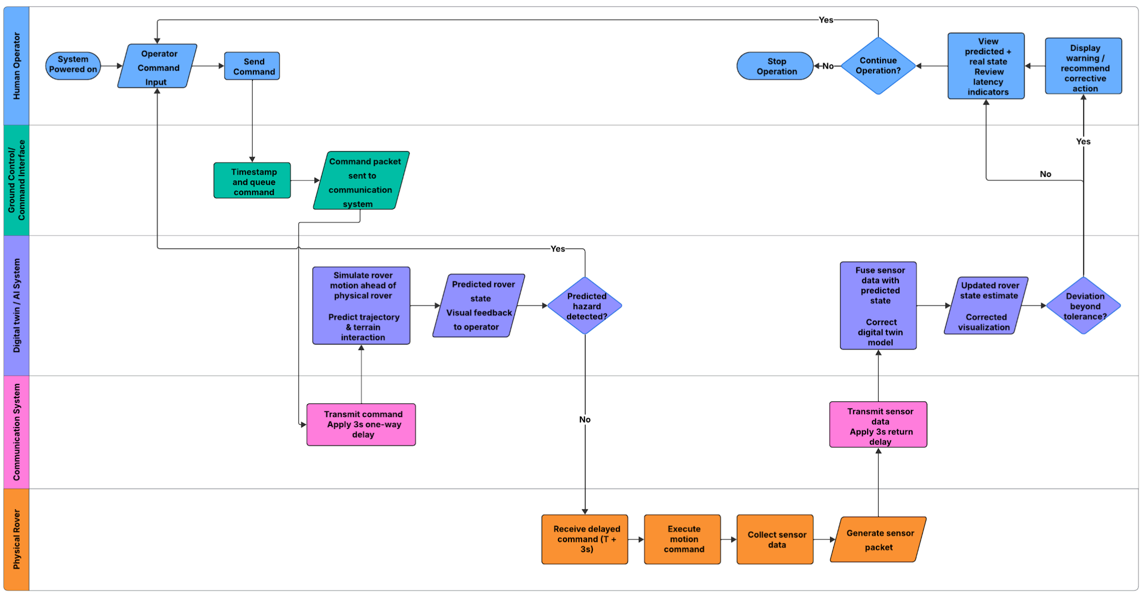

Figure 5: Command Process Map

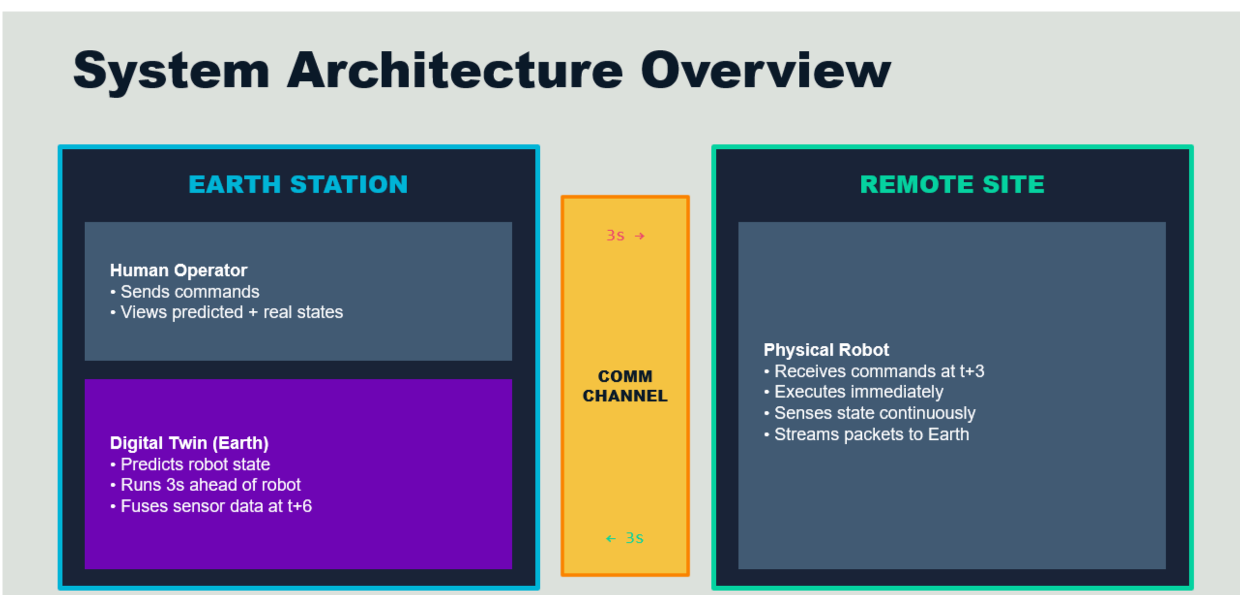

Figure 6: Overview of Command System

Human Operator Timeline (Earth)

Time	Human Operator: Actions & Display
t=0	Sends CMD_MOVE. UI shows "Command sent; one-way delay 3s"
t=1	Sees digital twin predicted motion (future state). No real sensor feedback yet.
t=2	Continues monitoring predicted state; can queue another command if desired.
t=3	Still watching predicted state; robot just received command remotely (no confirmation yet).
t=4	Digital twin continues predicting; human may adjust plan based on prediction.
t=5	Same; human can issue CMD_STOP or CMD_TURN if prediction shows risk.
t=6	First real sensor packets arrive (measured at robot-time t=3). Reality overlays prediction.
t=7	Receives sensor packets from robot-time t=4. Twin correction becomes visible.
t=8	Receives sensor packets from robot-time t=5. Prediction confidence improves.
t=9	Receives sensor packets from robot-time t=6. System is now well aligned.

Command Sent Prediction Phase Real Data Arriving

Figure 7: 9 second Timeline, as viewed by Operator

Digital Twin Timeline (Earth)

Time	Digital Twin: Compute & Update Behavior
t=0	Receives command instantly, begins simulation and prediction.
t=1	Predicts rover state corresponding to when command will reach robot (t=3) and beyond.
t=2	Continues predicting (nominally stays ~3s ahead of robot's actual state).
t=3	Continues predicting; robot is starting execution remotely now.
t=4	Continues predicting; flags risk if predicted slip/obstacle etc.
t=5	Continues predicting; can incorporate additional human commands instantly.
t=6	Receives first sensor packets (from robot-time t=3) → performs data fusion and corrects model.
t=7	Receives sensor packets (robot-time t=4) → refines state estimate and prediction.
t=8	Receives sensor packets (robot-time t=5) → reduces uncertainty further.
t=9	Receives sensor packets (robot-time t=6) → twin is now strongly synchronized with reality.

Initialize Pure Prediction Data Fusion & Correction

Figure 8: 9 second Timeline, as viewed by Digital Twin

Physical Robot Timeline (Remote Site)

Time	Execution State	Sensing State	Sending State
t=0	Idle (no command yet)	Sensors may be on (heartbeat)	No command-related telemetry
t=1	Idle	Same	Same
t=2	Idle	Same	Same
t=3	CMD arrives → Execution starts!	Measures state at t=3	Sends S(3) → arrives Earth t=6
t=4	Executing motion	Measures state at t=4	Sends S(4) → arrives Earth t=7
t=5	Executing motion	Measures state at t=5	Sends S(5) → arrives Earth t=8
t=6	Executing motion	Measures state at t=6	Sends S(6) → arrives Earth t=9
t=7	Executing motion	Measures state at t=7	Sends S(7) → arrives Earth t=10
t=8	Executing motion	Measures state at t=8	Sends S(8) → arrives Earth t=11
t=9	Execution finishing	Measures state at t=9	Sends S(9) → arrives Earth t=12

S(n) = Sensor packet measured at robot-time n, sent immediately, arrives at Earth at time n+3

Figure 9: 9 second Timeline, as viewed by physical robotmeline, as viewed by physical robot

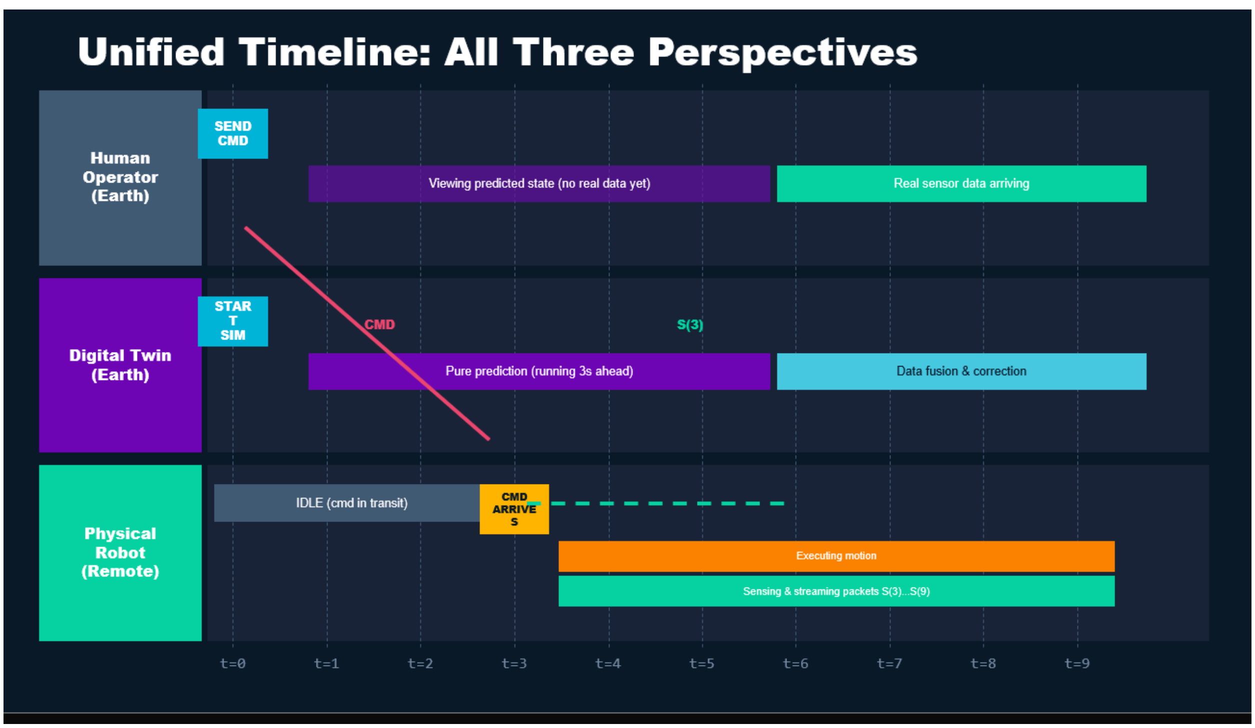

Figure 10: 9 second timeline combined viewline combined view